



YEAR 7 MATHEMATICS · CHAPTER 10

The median and range

Lesson 6 of 8

**In this lesson you will learn to:**

- find the median and range of a set of data
- find the median when there are two middle values
- use the median and range to compare and describe data

Name

Class

Teacher

Before you start

You will need

Pencil

Calculator

Syllabus: MA4-DAT-C-02: analyses simple datasets using measures of centre, range and shape of the data

WE DO

Grey boxes. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box.

- 1 Work through **Level 1**, then **Level 2**, then **Level 3**. Try the **extension** if you finish early.

- 2 At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

Lesson map

Page	What is on it	Done
3	Learning intentions, key terms, notes and Level 1 teacher examples	☐
4	Level 1: YOU DO practice questions	☐
5	Level 1 and 2: WE DO teacher examples and YOU DO practice questions	☐
6	Level 2: YOU DO practice questions	☐
7	Level 3: WE DO teacher examples and YOU DO practice questions	☐
8	Level 3: YOU DO practice questions	☐
9	Extension questions and the tear-off exit ticket	☐
10	Lesson summary: a worked example for every level	☐

YOU DO

On your own: Warm-up. These skills will help you in this lesson. Show your working.

- 1 Order: 9, 2, 7, 4, 6

- 3 $20 - 5$

- 2 Halfway between 7 and 9

- 4 $8 - (-3)$



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.06

The median and range

Date _____

Learning intentions

We are learning to:

- find the median and range of a set of data
- find the median when there are two middle values
- use the median and range to compare and describe data

Success criteria

I can:

- ☐ put data **in order** and find the middle value
- ☐ find the median of an even number of values
- ☐ find the **range** by subtracting the lowest value from the highest

Key terms Write the definition as your teacher explains it.

Median

Range

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Find the **median** of: 3, 9, 4, 7, 4, 8, 5

Example 2 Find the **range** of: 12, 5, 9, 20, 7

YOU DO

On your own: Set A. Find the median. Put the numbers in order first!

1 2, 3, 3, 5, 8**6** 4, 9, 7, 9, 2, 5, 9**2** 1, 4, 4, 6, 9**7** 15, 13, 14, 13, 18**3** 7, 2, 5, 2, 9**8** 8, 3, 6, 3, 1, 5, 3**4** 6, 1, 8, 6, 3**9** 20, 25, 22, 25, 21**5** 10, 12, 11, 10, 15**10** 5, 7, 6, 7, 9, 4, 8

YOU DO On your own: Set B. Find the range (highest – lowest).

1 3, 8, 5, 10

6 6, 6, 6, 6

2 14, 6, 23, 9, 11

7 45, 12, 33, 50, 19

3 1, 1, 6, 4

8 7.5, 3.2, 6.1

4 25, 30, 18, 22

9 0, 15, 9, 3

5 100, 85, 92, 97

10 -2, 5, 1, 3

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the median and range of: 8, 3, 10, 5, 7, 12**Example 4** Find the median and range of the data in this stem-and-leaf plot.

Stem	Leaf
2	4 8
3	1 5 5 9
4	2

Key: 3 | 1 = 31

YOU DO

On your own: Set C. Find the median and the range.

1 4, 9, 2, 7**2** 11, 15, 12, 18, 14, 20**3** 3.5, 1.2, 4.8, 2.6, 3.1**4** 9, 2, 6, 11, 4, 8**5** Temperatures ($^{\circ}\text{C}$): -4, 2, -1, 5, 0, 3**6** Temperatures ($^{\circ}\text{C}$) for a week: 21, 24, 19, 24, 22, 23, 21**7** Scores: 1 three times, 2 four times, 3 twice, 4 once**8** Heights (cm): 150, 145, 160, 155, 148, 152**9** Prices (\$): 100, 250, 175, 300, 125**10** Goals: 0 twice, 1 four times, 2 three times, 3 once



Level 3 · Challenge

WE DO

With your teacher: copy what they write on the board.

Example 5

Ages at a party: 11, 12, 12, 13, 12, 44.
Find the median, range and mean.
Which describes a typical age best?

Example 6

Write five numbers with a median of 8
and a range of 10. The smallest is 3.

YOU DO

On your own: Set D. Show your working.

- 1 Write a set of five numbers with a median of 6 and a range of 8.

- 3 The number 20 is added to 4, 6, 7, 8, 10. How do the median and range change?

- 2 The range of a data set is 15 and the smallest value is 7. What is the largest value?

- 4 Class A: median 65, range 40. Class B: median 60, range 10. Which class is more consistent? Which did better?

YOU DO On your own: Set D continued. Show your working.

- 5 Explain why one very large value changes the mean a lot but the median only a little.

- 8 Heights (cm): 180, 175, 190, 182 and one more. The median is 182 and the range is 20. Find the missing height.

- 6 Six numbers have a median of 10. Four are 4, 8, 13 and 15. The other two are equal. What are they?

- 9 Can the range of a data set be 0? Explain.

- 7 Write five numbers with median 8, range 10, smallest 3 and mean 8.

- 10 Ages in a family: 42, 40, 14, 11, 8, 1. Find the median and range. Does the median describe this family well?

Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

25 values are written in order. Which value is the median?

1. Add 1 to the number of values: $25 + 1 = 26$
2. Halve it: $26 \div 2 = 13$
3. The median is the 13th value (12 values sit on each side)

Key idea: the median is in position $(n + 1) \div 2$. If that ends in .5, average the two values either side.



Answer: the 13th value

YOU DO

On your own: read the worked example, then try these.

- E1** There are 40 values in order. Which two positions are used to find the median?

- E2** Scores: 1 three times, 2 five times, 3 eight times, 4 four times. Find the median.

- E3** Find the median of all the whole numbers from 1 to 100.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the median of: 9, 3, 7, 5, 11

Level 2

Find the median and range of: 9, 3, 7, 3, 11, 6

Level 3

Write five numbers with a median of 6 and a range of 10. The smallest is 2.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Put the data **in order** first.
 2. **Median** = the middle value.
 3. Two middle values? Add them and **divide by 2**.
 4. **Range** = highest – lowest.

Level 1

Find the median of 7, 3, 9, 5, 4

1. Order: 3, 4, 5, 7, 9
2. Middle (3rd) value: 5

Answer: 5**Level 2**

Find the median and range of 6, 2, 9, 4

1. Order: 2, 4, 6, 9
2. Two middle values: $(4 + 6) \div 2 = 5$
3. Range: $9 - 2 = 7$

Answer: median 5, range 7**Level 3**Temperatures ranged from -3°C to 8°C . Find the range.

1. Range = highest – lowest
2. $8 - (-3) = 8 + 3 = 11$

Answer: 11°C **Extension**

Where is the median of 31 ordered values?

1. $(31 + 1) \div 2 = 16$

Answer: the 16th value

This space is left blank on purpose. It is the back of your exit ticket.